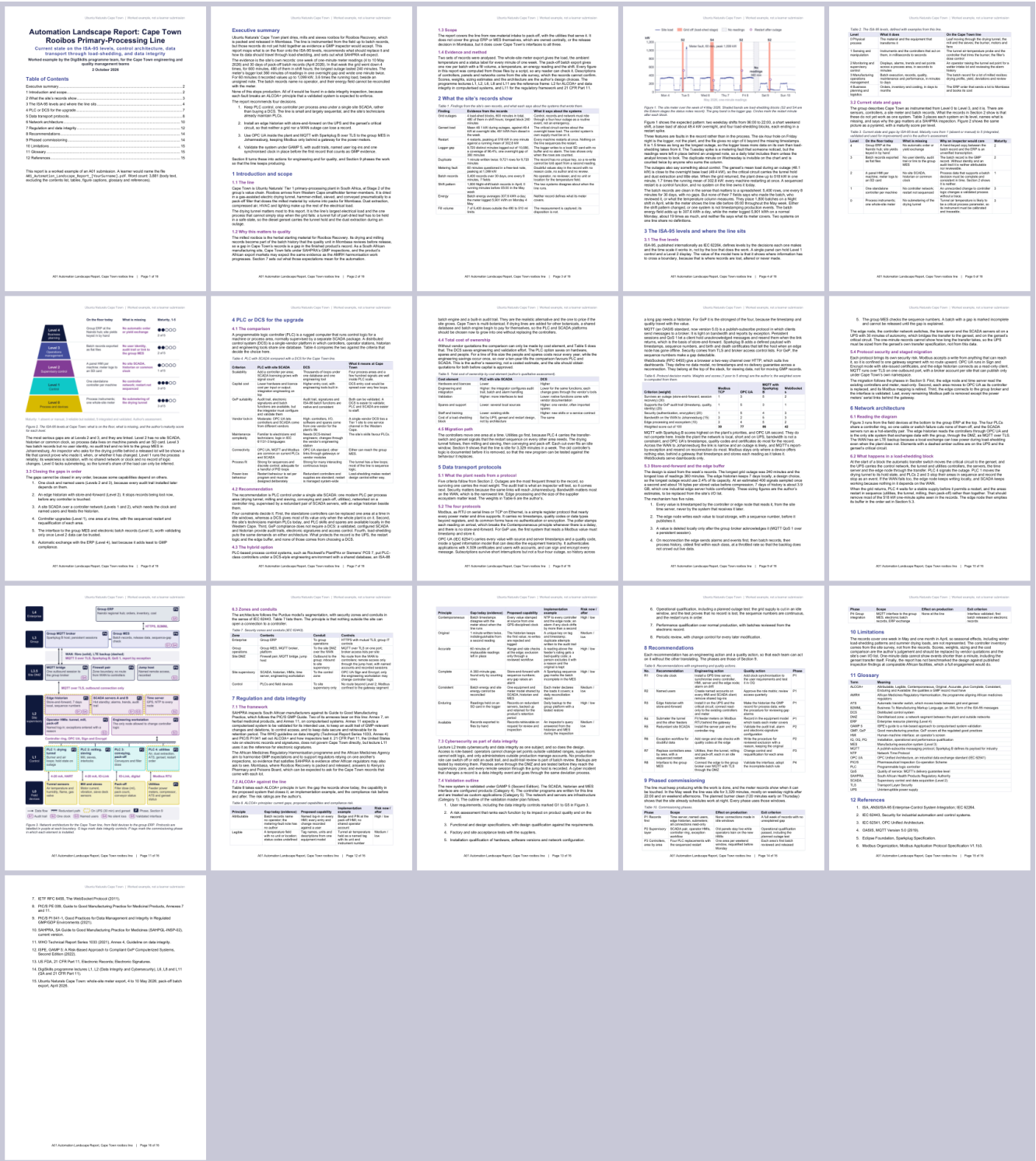
A01 worked exemplar: automation landscape report for the Cape Town rooibos line

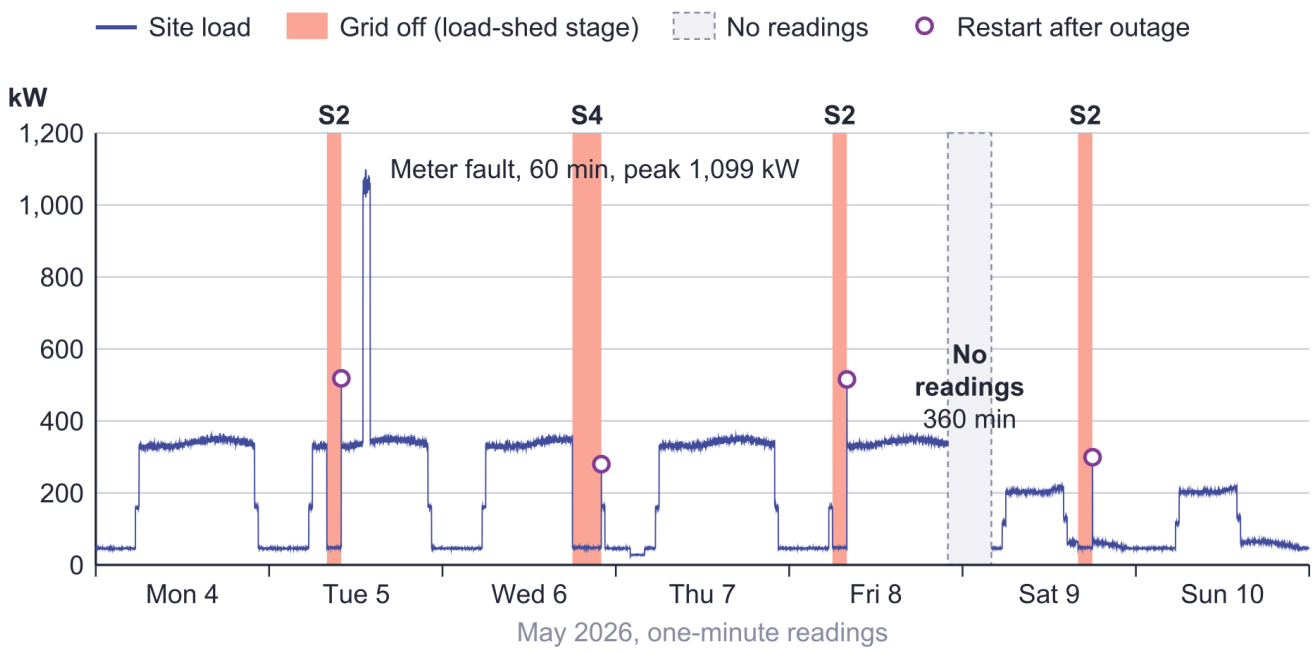
This is a worked example of an A01 submission, written by the programme team to show a technical report at the upper bands. It is set at Ubuntu Naturals' Cape Town rooibos primary-processing line, and not at the Johannesburg encapsulation plant your brief describes. The two problems differ at the root. Cape Town dries, mills and sieves a herbal starting material, holds a thermal process on a genset through load-shedding, and sends its records to Mombasa for release; Johannesburg runs a capsule line on legacy standalone PLCs and is the group's MES anchor. Do not copy it. The upper bands are awarded for fitting the pyramid, the controller choice, the protocols and the ALCOA+ mapping to Johannesburg's equipment and gaps, and borrowed findings or wording answer questions about the wrong plant.

The submission

The submission is `A01_exemplar_cape_town_rooibos_report.pdf`, 16 A4 pages, with the editable `A01_exemplar_cape_town_rooibos_report.docx` it was exported from. A learner names the same file `A01_Automation_Landscape_Report_[YourSurname].pdf`. Body text is Arial 11 pt, sections are numbered, every page carries a page number, and the body runs to 3,881 words against the brief's 3,000 to 4,000 (tables, captions, glossary and references excluded, which the report states on its first page). The contact sheet below shows every page; read the PDF at 100%.

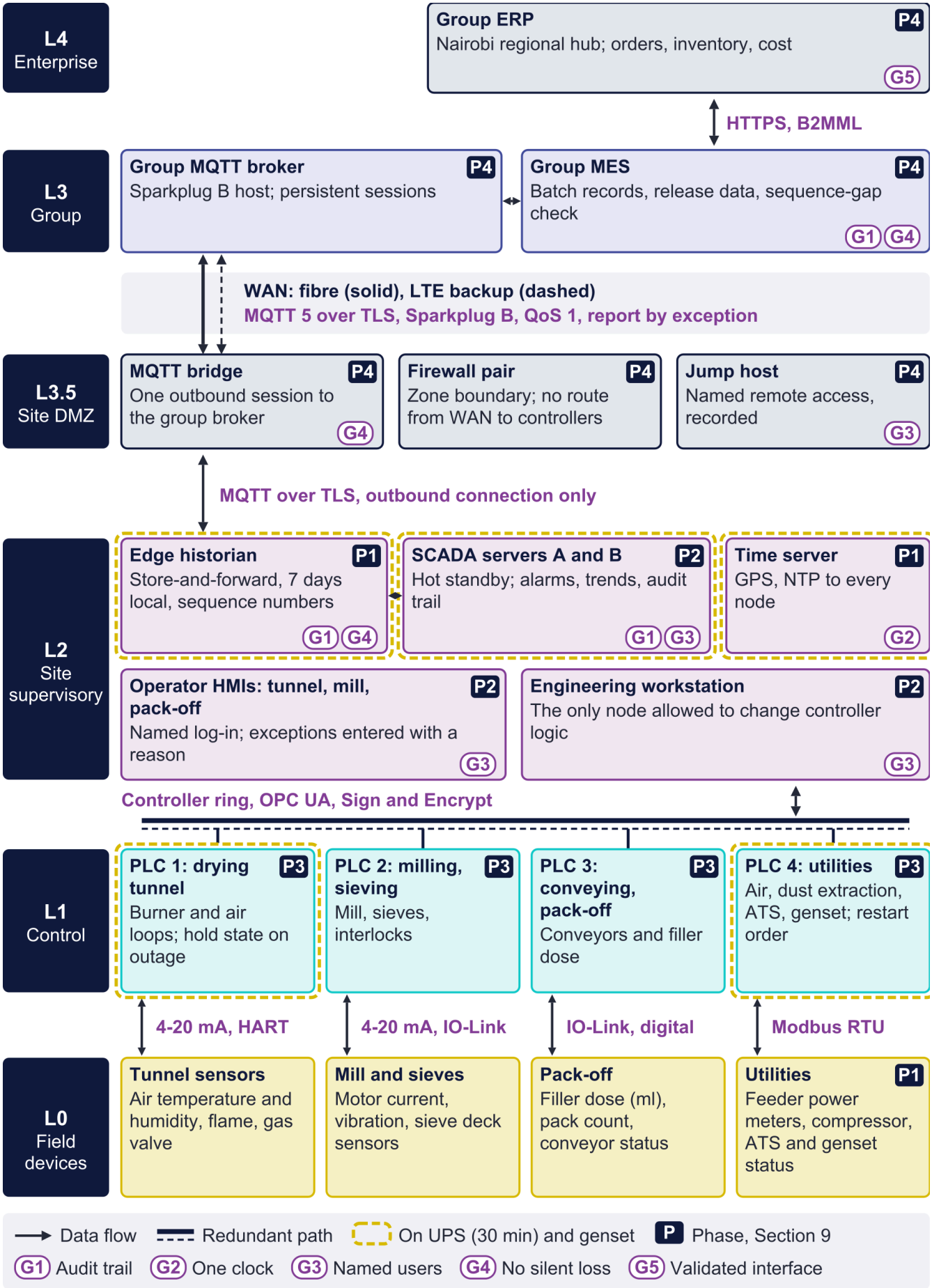


The three figures are authored at the 16 cm width they print at, so no label prints smaller than 9 pt. Figure 1 is the evidence the report rests on, Figure 2 is the pyramid mapping, and Figure 3 is the architecture.



	On the floor today	What is missing	Maturity, 1-5
Level 4 Business planning	Group ERP at the Nairobi hub; site yields keyed in by hand	**No automatic order or yield exchange**	●●○○○ 2 of 5
Level 3 Operations management	Batch records exported as flat files	**No user identity, audit trail or link to the group MES**	●●○○○ 2 of 5
Level 2 Supervisory control	A panel HMI per machine; meter logs to an SD card	**No site SCADA, historian or common clock**	●○○○○ 1 of 5
Level 1 Control	One standalone controller per machine	**No controller network; restart not sequenced**	●●●○○ 3 of 5
Level 0 Process and devices	Process instruments; one whole-site meter	**No submetering of the drying tunnel**	●●●○○ 3 of 5

Maturity: 1 absent or manual, 3 reliable but isolated, 5 integrated and validated. Author's assessment.


L4
Enterprise
Group ERP
P4
Nairobi regional hub; orders, inventory, cost
G5
HTTPS, B2MML
L3
Group
Group MQTT broker
P4
Sparkplug B host; persistent sessions
Group MES
P4
Batch records, release data, sequence-gap check
G1 G4
WAN: fibre (solid), LTE backup (dashed)
MQTT 5 over TLS, Sparkplug B, QoS 1, report by exception
L3.5
Site DMZ
MQTT bridge
P4
One outbound session to the group broker
G4
Firewall pair
P4
Zone boundary; no route from WAN to controllers
Jump host
P4
Named remote access, recorded
G3
MQTT over TLS, outbound connection only
L2
Site supervisory
Edge historian
P1
Store-and-forward, 7 days local, sequence numbers
G1 G4
SCADA servers A and B
P2
Hot standby; alarms, trends, audit trail
G1 G3
Time server
P1
GPS, NTP to every node
G2
Operator HMIs: tunnel, mill, pack-off
P2
Named log-in; exceptions entered with a reason
G3
Engineering workstation
P2
The only node allowed to change controller logic
G3
Controller ring, OPC UA, Sign and Encrypt
L1
Control
PLC 1: drying tunnel
P3
Burner and air loops; hold state on outage
PLC 2: milling, sieving
P3
Mill, sieves, interlocks
PLC 3: conveying, pack-off
P3
Conveyors and filler dose
PLC 4: utilities
P3
Air, dust extraction, ATS, genset; restart order
4-20 mA, HART
4-20 mA, IO-Link
IO-Link, digital
Modbus RTU
L0
Field devices
Tunnel sensors
Air temperature and humidity, flame, gas valve
Mill and sieves
Motor current, vibration, sieve deck sensors
Pack-off
Filler dose (ml), pack count, conveyor status
Utilities
P1
Feeder power meters, compressor, ATS and genset status
Data flow
Redundant path
On UPS (30 min) and genset
P Phase, Section 9
G1 Audit trail
G2 One clock
G3 Named users
G4 No silent loss
G5 Validated interface

Start from the site's own records

The report does not open with a textbook pyramid. It opens with what the site's records show: a week of one-minute meter readings and a month of batch records. Those two files carry every finding the later sections act on. The grid failed 4 times for 600 minutes, the longest block lasted 240 minutes, the logger lost 360 minutes on its own, one minute was written twice, and 60 minutes of doubtful readings stayed in the record behind an unsigned note. The batch file has no operator field, places 1,800 batches on a Night shift that the meter does not show, and reports an energy figure that cannot be reconciled with the meter. Your brief gives you no data file for Johannesburg, so your equivalent is the survey of the capsule line you are asked to describe; the habit to copy is grounding each claim in something the reader can check.

Let the evidence set the design

Each design choice in the report answers a finding. The edge buffer is sized against the longest outage and the longest gap. The sequenced restart answers the 518 kW one-minute spike. The UPS and genset coverage follows from the genset carrying about the overnight base load. The ALCOA+ table fills its "gap today" column from the same records, so each principle points at a real fault rather than a generic risk. A marker reads this as fit to the plant, which is what the Distinction descriptors ask for.

Keep data and design apart

The report says which numbers are measured and which are the author's: the decision-matrix weights and scores, the maturity scores, the 400-signal estimate, the 7-day retention and the 30-minute UPS autonomy are all labelled as design choices, and the cost comparison is qualitative because there are no quotations. Where it relies on the site survey (one controller per machine, a panel HMI on each), it says so in Section 1.4 and again in the limitations.

Write for two readers

The executive summary stands on its own for the quality manager. The recommendations table gives every action an engineering column and a quality column, so neither team has to translate the other's. Acronyms are defined on first use and again in the glossary.

How it maps to the marking scheme

Rubric row	Band it aims at	Why	What would lift it a band
ISA-95 Pyramid Mapping	Distinction	Every level is defined with a line-specific example, every existing system is placed, and every gap is named with the reason it matters at a SAHPRA inspection. It also scores maturity per level and orders the gap closure by dependency, which reaches into Stretch.	Benchmark each level against the public record for comparable African facilities (WHO prequalification inspection reports, published SAHPRA findings), which Stretch requires and the report does not attempt.
PLC vs. DCS Comparison	Distinction	Eight criteria, a clear choice, and a justification built on the site's constraints: area-by-area migration in idle windows, local PLC skills, GxP met by a validated SCADA, and load-shedding being architecture-neutral. A hybrid option and future multi-botanical lines are discussed, and the cost of ownership is reasoned element by element.	Real vendor specifications and a peer case study, and a costed ten-year comparison in place of the qualitative cost table.
Protocol Evaluation	Stretch	Four protocols are tested against the plant's constraints with a weighted decision matrix, a store-and-forward design sized from the outage data, the audit-trail consequence of each protocol, protocol-level security and a staged migration. It holds the band narrowly, because the scores are the author's judgement with no cited test or vendor data behind them.	Ground the scores in sources: published payload and bandwidth figures, and a pilot that pulls the grid and counts the sequence numbers that come back.
Network Architecture Diagram	Distinction	The diagram runs from field devices to ERP with a protocol at each boundary, a redundant controller ring and SCADA pair, dual WAN paths, UPS coverage for a load-shedding block, data integrity tags and commissioning phases. The zones and conduits are tabulated in Section 6.3 rather than drawn.	A companion physical view (panels, cable routes, power circuits) and the IEC 62443 zones drawn as boundaries on the diagram itself.

Rubric row	Band it aims at	Why	What would lift it a band
Regulatory and Data Integrity (ALCOA+)	Distinction	Each ALCOA+ principle is mapped to a gap the records show today, a proposed capability and an implementation example, with risk before and after. SAHPRA, PIC/S Annexes 7 and 11, WHO TRS 1033 Annex 4 and the AMRH context are placed, and L2 cybersecurity is treated as part of data integrity.	Cite published inspection findings at comparable facilities, score the risk numerically, and expand the validation outline into a fuller validation master plan with deliverables per stage.
Communication Quality	Distinction	A standalone executive summary, numbered sections, a contents list, a glossary, captioned figures and tables, and recommendations with separate engineering and quality actions. The prose is plain and the data is checkable.	A one-page visual summary for the board, and a lighter Section 5, whose protocol descriptions run longer than a quality manager needs.

Where the figures come from

Every data figure in the report is computed by `scripts/build_a01_exemplar.py` from the files below, and `tests/test_exemplar_a01.py` resolves each locator again with its own code and checks that the DOCX and PDF carry the value. In the locators, `count[...]`, `runs[...]` and `longest_run[...]` filter rows by column value (`~` means "contains"), `shift_hours` is the shift pattern the W09 asset README states (weekdays 06:00 to 22:00, weekends 06:00 to 14:00), and `hour<5` is a timestamp before 05:00. The 7-day meter file is the earlier seven-day edition of the W09 dataset, kept beside the current workshop. The design choices (weights, scores, retention, signal count, UPS autonomy, the architecture) are the author's and have no source.

Figure	Value	Source
Meter rows in the file	9,721	`workshops/09 energy monitoring/graduate/group assets/cape_town_rooibos_energy_7d.csv:rows`
Distinct minutes logged	9,720	`workshops/09 energy monitoring/graduate/group assets/cape_town_rooibos_energy_7d.csv:distinct(timestamp)`
Minutes in the week	10,080	`workshops/09 energy monitoring/graduate/group assets/cape_town_rooibos_energy_7d.csv:span_minutes(timestamp)`
Coverage	96.4%	`workshops/09 energy monitoring/graduate/group assets/cape_town_rooibos_energy_7d.csv:coverage_pct(timestamp)`

Figure	Value	Source
Longest logging gap, min	360	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:longest_gap_minutes(timestamp)`
Duplicated minutes	1	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:duplicates(timestamp)`
Load-shed minutes	600	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:count[line_status=load_shed]`
Load-shed blocks	4	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:runs[line_status=load_shed]`
Longest block, min	240	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:longest_run[line_status=load_shed]`
Load-shed minutes in shift hours	480	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:count[line_status=load_shed & shift_hours]`
Mean genset load, kW	48.1	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:mean(load_kw)[line_status=load_shed]`
Diesel-supplied energy, kWh	481	`workshops/09 energy monitoring/graduate/group assets/cape town rooibos energy 7d.csv:sum(load_kw)/60[line_status=load_shed]`
Mean idle load, kW	49.4	`workshops/09 energy monitoring/graduate/group assets/cape town rooibos_energy_7d.csv:mean(load_kw)[line_status=idle]`
Idle minutes	3,329	`workshops/09 energy monitoring/graduate/group assets/cape_town_rooibos_energy_7d.csv:count[line_status=idle]`
Mean running load, kW	302.8	`workshops/09 energy monitoring/graduate/group assets/cape town rooibos_energy_7d.csv:mean(load kw)[line_status=running & notes!~fault]`

Figure	Value	Source
Restarts	4	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:count[line_status=restart]`
Peak restart load, kW	518	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:max(load_kw)[line_status=restart]`
Metering-fault minutes	60	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:count[notes~metering fault]`
Peak metering-fault reading, kW	1,099	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:max(load_kw)[notes~metering fault]`
Maintenance minutes	120	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:count[line_status=maintenance]`
Running minutes before 05:00	0	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:count[line_status=running & hour<5]`
Energy on Monday 4 May, kWh	5,901	`workshops/09_energy_monitoring/graduate/group_assets/cape_town_rooibos_energy_7d.csv:sum(load_kw)/60[date=2026-05-04]`
Batch records	5,400	`workshops/02 analytical tools/graduate/group assets/W2 files/cape_town_rooibos_30d.csv:rows`
Batch days	30	`workshops/02 analytical tools/graduate/group assets/W2 files/cape_town_rooibos_30d.csv:days(Timestamp)`
Batch interval, min	8	`workshops/02 analytical tools/graduate/group assets/W2 files/cape_town_rooibos_30d.csv:interval_minutes(Timestamp)`
Batch record fields	7	`workshops/02 analytical tools/graduate/group assets/W2 files/cape_town_rooibos_30d.csv:columns`
Night-shift batch records	1,800	`workshops/02 analytical tools/graduate/group assets/W2 files/cape_town_rooibos_30d.csv:count[Shift=Night]`

Figure	Value	Source
Doses outside the fill limits	7	`workshops/02_analytical_tools/graduate/group_assets/W2_files/cape_town_rooibos_30d.csv:count_outside(Fill_Volume_ml,490,510)`
Batch energy per day, kWh	307.6	`workshops/02_analytical_tools/graduate/group_assets/W2_files/cape_town_rooibos_30d.csv:sum(Energy_kWh)/days(Timestamp)`
Lower fill limit, ml	490	`workshops/02_analytical_tools/graduate/LEARNER.md:re(LSL_Vol = (\d+))`
Upper fill limit, ml	510	`workshops/02_analytical_tools/graduate/LEARNER.md:re(USL_Vol = (\d+))`